

IAL Physics Unit 3 (WPH13): Rapid Revision

Practical Skills in Physics I · last-minute summary · 1 h 20 min · 50 marks · 4 practical questions · no MCQ, no 6-mark answer

HOW THE MARKS WORK

- Most 2-mark answers are an **action + reason** pair: “check for zero error → to eliminate systematic error”. Always write both halves.
- Mean: same d.p. as the data. Calculated column: 3 s.f. Constant from a gradient: 2–3 s.f. **with unit**. % uncertainty: 1–2 s.f.
- Conclusions need a number, a comparison and a decision.

INSTRUMENTS

Metre rule	1 mm
Vernier calliper	0.1 mm (no parallax, fits curves)
Digital calliper / micrometer	0.01 mm (micrometer: use the ratchet)
Protractor	1°
Stopwatch	0.01 s, but reaction time ≈ 0.2 s
Balance	1 g / 0.1 g / 0.01 g (tare first)
Digital meter / thermometer	last digit; no parallax
CRO	$t = \text{divisions} \times \text{time base}$
Light gates + logger	no reaction time; must be aligned

UNCERTAINTIES

Single reading	half the resolution (full resolution usually scores 1 of 2)
Repeats	mean (drop a clear anomaly); half the range (or furthest from the mean)
% U	$\text{uncertainty} \div \text{value} \times 100$
Many items	measure 20 together: same uncertainty, bigger reading, smaller % U
Combining	$\times$ and $\div$ add %; a power multiplies it ($r^3 \rightarrow 3 \times$)

COMPARE WITH AN ACCEPTED VALUE

Limit: *your* value $\times (1 \pm \%)$, then is the accepted value inside the range?

% difference: $|\text{accepted} - \text{yours}| \div \text{accepted} \times 100$, then compare with the % uncertainty. Example: $8.65 \times 1.009 = 8.73 < 8.94$, so not copper. Using the limit of the accepted value scores 0.

ERRORS

- **Random** (scatter: reaction time, parallax, judging a maximum): reduce by repeats and a mean, timing longer intervals, many points and a best-fit line. It cannot be eliminated.
- **Systematic** (all shifted: zero error, background light, warped rule): repeats do not help; check the zero, calibrate, control. Shows as an intercept, so a gradient ignores it.
- **Precise** = small spread; **accurate** = close to the true value.
- **Why a graph:** more pairs, averaging, anomalies visible, systematic error removed.

CRITICISE THE TABLE (PICK 2-3)

- Inconsistent d.p. (measured) or s.f. (calculated).
- Not recorded to the instrument’s resolution.
- No repeats (only if none are shown).
- Too few readings ($< 5-6$); range too small; uneven gaps.
- A named value off the trend, or wrongly rounded.
- Heading missing “quantity / unit”.

GRAPH MARKS

1. **New column** to 3 s.f., heading like “ $1/\lambda / 10^6 \text{ m}^{-1}$ ”.
 2. **Axes** “quantity / unit”, correct way round; $\sin \theta$ and n have no unit.
 3. **Scales** 1, 2, 5 per 2 cm; points cover $\geq$ half the grid each way.
 4. **Plots** within 1 mm (one error loses the mark).
 5. **Best-fit line:** thin, single, balanced; not forced through the origin.
- **Gradient:** triangle $>$ half the line, values **from the line**, show the working.
 - **Proportional:** straight line **and** through the origin. Inverse: plot against $1/x$ or check xy is constant.
 - **Linearise:** rearrange, line up under $y = mx + c$, then “gradient = ..., constants ...”.

LINEARISATIONS SEEN

λ vs $\sin \theta$	gradient d
$\sin \theta_2$ vs $\sin \theta_1$	gradient n
V (or E) vs $1/\lambda$	gradient hc/e (or hc)
λ vs $1/\nu$	gradient h/m
I vs $1/d^2$	gradient k , through the origin
R vs l	gradient ρ/A
R/L vs L	intercept $\times A = \rho$
R vs θ	α = gradient/intercept
V vs I	intercept ε , gradient $-r$
F vs N	gradient μ
d vs F (beam)	$E = l^3/(4wh^3 \times \text{gradient})$
f vs n	$v = 2L \times \text{gradient}$
v^2 vs d	gradient $k \approx g$
$1/t^2$ vs h	$g = 3s^2 \times \text{gradient}$
m vs v^2	$C = \text{gradient} \times 2g/(\rho A)$

TECHNIQUES (ACTION → REASON)

Set square / eye level / rule close	avoid parallax
Spirit level, set square to bench	rule vertical
Check zero (callipers, micrometer, balance)	eliminate systematic error
Micrometer: several places + orientations	mean; non-uniform object
Time many cycles / long distance	reaction time a smaller fraction
Light gates, logger, video	remove reaction time
Equal sections, equal times	terminal velocity reached
Approach resonance from both sides	mean frequency
Dark room / cover / subtract background	background light
Monochromatic light (laser, LED)	single λ , sharp angle
Protective resistor; variable resistor at max	limit current, no overheating
Small current / switch off between readings	wire does not heat up
Water bath + stir + thermometer close	known, uniform temperature

SAFETY (HAZARD → CONTROL)

Masses falling	feet clear; box/sand tray; boots
Wire/band/cord snapping	safety goggles (not "glasses")
Stand toppling	clamp to bench / heavy mass on base
Laser	never look into the beam; avoid reflective surfaces
Hot water	clamp the beaker; heat-proof gloves
Loud sound	ear defenders; limit volume and time
Glass breaking	use the ratchet; gloves + goggles

PRACTICALS IN ONE LINE EACH

- **Density:** balance (many items) + callipers/micrometer means $\rightarrow V \rightarrow \rho$ (3 s.f.) $\rightarrow$ % range vs the table.
- **k / Young modulus:** $F = mg$; extension with pointer + set square; long thin wire; gradient of the linear part; goggles.
- **Viscosity:** bands below the surface; equal times; $\eta = (W - U)/6\pi r v$; constant T .
- **g :** light gate + card, $v^2 = 2as$; or s against t^2 .
- **Speed of sound:** tube $\lambda = 4L$; echo $v = 2d/t$; CRO time base.
- **String:** $\mu = mg/(f\lambda)^2$; rule held against the string.
- **Refractive index:** normal drawn; ≥ 5 angles both sides; $\sin$ vs $\sin$ gradient.
- **Resistivity:** R against l ; $\rho = \text{gradient} \times A$; intercept = contact resistance.
- **e.m.f./ r :** V against I ; protective resistor.
- **Planck (LEDs):** V_a against $1/\lambda$; LEDs emit a band of λ , so uncertainty in h .
- **Efficiency:** useful $mgh/t \div$ input VI .
- **Inverse square:** measure to the filament (add the bulb radius); I against $1/d^2$.